

1. An instructor gives the same data as given below to four students: (1, 1), (10, 100), (20, 400), (30, 600), (40, 1200) They each come up with four different answers for the straight line regression model. Only one is correct. The correct model is Select one:

- a) $y = 30x - 200$
 - b) $y = -139.43 + 29.684x$
 - c) $y = 60x - 1200$
 - d) $y = 1 + 22.782x$
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2. With a Kohonen network (SOM), the output layer node that wins an input instance is rewarded by having: Select one:

- a) neighboring connection weights modified to become less similar to its own connection weights.
 - b) its connection weights modified to more closely match those of its neighbors.
 - c) a higher probability of winning the next training instance to be presented.
 - d) its connection weights modified to more closely match those of the input instance.
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3. In machine learning the simple linear regression equation can be written as $y = b + wx$ the variable y represents the Select one:

- a) average or predicted response
 - b) independent variable
 - c) estimated weight
 - d) estimated bias
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4. Transfer learning is: Select one:

- a) the machine learning task of learning a function that maps an input to an output based on example input-output pairs.
 - b) a machine learning method where a model developed for a task is reused as the starting point for a model on a second task
 - c) an area of machine learning concerned with how software agents ought to take actions in an environment in order to maximize the notion of cumulative reward.
 - d) a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labeled responses.
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5. What is the name of this phenomenon that sometimes occurs in machine learning model: "occurs when the training error of the learnt model is small but the test error is large"? Select one:

- a) Overfitting
- b) Generalization
- c) Underfitting
- d) Classification

6. In neural network, one disadvantage of sigmoid activation function : Select one:

- a) it has a gradient vanishing phenomenon (its derivative is close to 0 at the point away from the central point)
- b) It is monotonic
- c) easy to derive
- d) It is continuous

7. Why are linearly separable problems interesting to neural network researchers? Select one:

- a) Because they are the only mathematical functions that you can draw.
- b) Because they are the only mathematical functions that are continuous.
- c) Because they are the only problems that a perceptron can solve successfully.
- d) Because they are the only problems that a neural network can solve successfully.

8. A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear i.e. ($f(x)=2 \cdot x$). The inputs are 4, 10, 5 and 20 respectively. The output will be: Select one:

- a) 238
- b) 123
- c) 119
- d) 76

9. A regression between foot length (response variable in cm, y) and height (independent variable in inches, x) for 33 students resulted in the following regression equation: $y = 10.9 + 0.23x$. One student in the sample was 73 inches tall with a foot length of 29 cm. What is the predicted foot length for this student? Select one:

- a) 27.69 cm

- b) 33 cm
- c) 17.57 cm
- d) 29 cm

